

METHODOLOGY REPORT

E-Commerce Conversion Prediction Challenge (Summer Analytics 2026)

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Selected Machine Learning Model - **Logistic Regression**

1. **Importing the required Libraries.**

2. **Loading the data.**

3. **Exploratory Data Analysis (EDA) – conducted to assess :**

- Description & Information of train data
- Missing values of **train.csv** & **private_test.csv**
- Target variable distribution
- Statistical Summary of **train.csv**

4. **Data Preprocessing & Feature Transformation :**

- Device_Type, Traffic_Source, and Campaign_Code - were converted into numerical features using **One-Hot Encoding and included** in the training.
- Imputed the Missing Value with the median value to prevent outlier bias.
- Features were normalized via **Standard Scaling** (StandardScaler) to achieve zero mean and unit variance.

5. **Model Development & Evaluation :**

- Predictive model strictly relies on a **Logistic Regression Classifier**.

By passing the imputed and scaled one-hot encoded dataset to the model.

6. **Conclusion & Output Generation :**

- The final model was used on the **private_test.csv** to generate an output file, submission.csv, with the columns **User_ID** and **Converted – Binary value (0,1)**, where 1 = Converted and 0 = Not Converted.